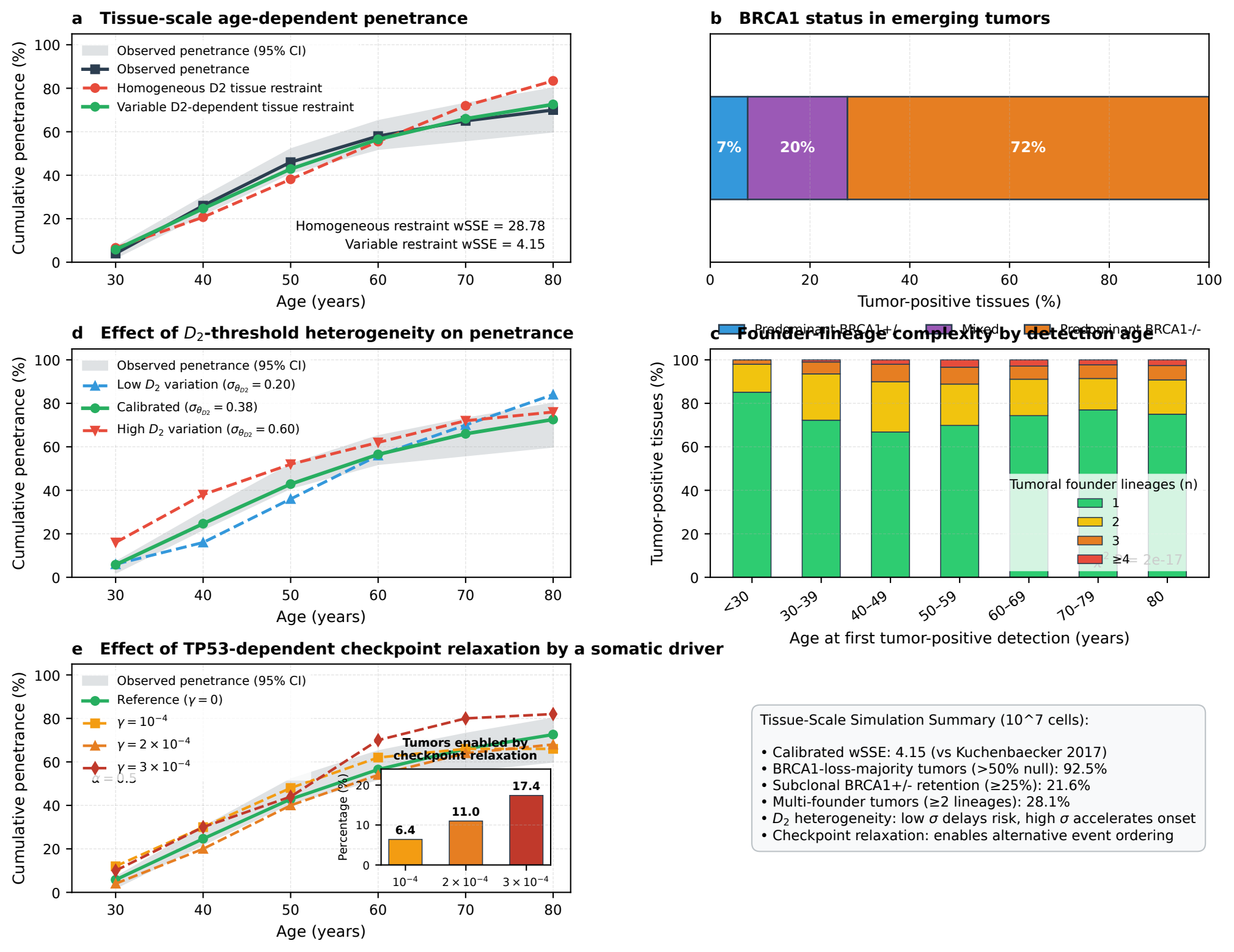
a Tissue-scale age-dependent penetrance
Observed penetrance (95% CI)
Observed penetrance
Homogeneous D2 tissue restraint
Variable D2-dependent tissue restraint
Homogeneous restraint wSSE = 28.78
Variable restraint wSSE = 4.15
Cumulative penetrance (%)
Age (years)
b BRCA1 status in emerging tumors
7%
20%
72%
Tumor-positive tissues (%)
d Effect of D_2-threshold heterogeneity on penetrance
Observed penetrance (95% CI)
Low D_2 variation ($\sigma_{\theta_{D2}} = 0.20$)
Calibrated ($\sigma_{\theta_{D2}} = 0.38$)
High D_2 variation ($\sigma_{\theta_{D2}} = 0.60$)
Cumulative penetrance (%)
Age (years)
c Founder-lineage complexity by detection age
Predominant BRCA1+/-
Mixed
Predominant BRCA1-/-
Tumoral founder lineages (n)
1
2
3
≥4
Tumor-positive tissues (%)
<30
30–39
40–49
50–59
60–69
70–79
80
Age at first tumor-positive detection (years)
e Effect of TP53-dependent checkpoint relaxation by a somatic driver
Observed penetrance (95% CI)
Reference ($\gamma = 0$)
$\gamma = 10^{-4}$
$\gamma = 2 \times 10^{-4}$
$\gamma = 3 \times 10^{-4}$
$\alpha = 0.5$
Tumors enabled by checkpoint relaxation
Percentage (%)
6.4
11.0
17.4
10^{-4}
2×10^{-4}
3×10^{-4}
Cumulative penetrance (%)
Age (years)
Tissue-Scale Simulation Summary (10^7 cells):
• Calibrated wSSE: 4.15 (vs Kuchenbaecker 2017)
• BRCA1-loss-majority tumors (>50% null): 92.5%
• Subclonal BRCA1+/- retention (≥25%): 21.6%
• Multi-founder tumors (≥2 lineages): 28.1%
• D_2 heterogeneity: low σ delays risk, high σ accelerates onset
• Checkpoint relaxation: enables alternative event ordering